RECTIFICATION

Rectification: The arc length S between two points $A(x_1, y_1)$ & $B(x_2, y_2)$ is

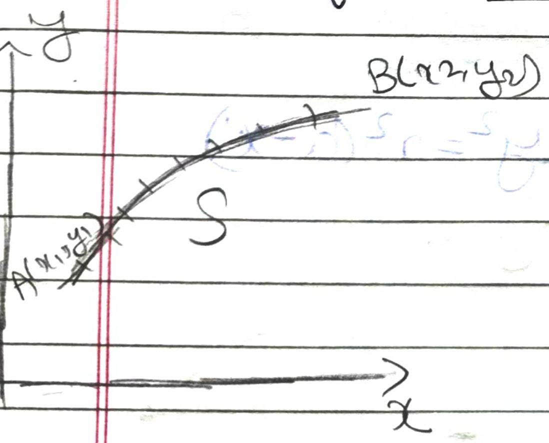
① $$\boxed{S = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx}$$ OR $$\boxed{S = \int_{y_1}^{y_2} \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy}$$

cartesian form.

② $$\boxed{S = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta \quad -②}$$ polar form

③ $$\boxed{S = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \quad -③}$$ parametric form

List of Standard eqns. in Cartesian form



① Straight line.

i) $y = mx$

ii) $y = mx + c$

iii) $\frac{x}{a} + \frac{y}{b} = 1$

iv) $ax + by = c$

3 → para 4 → ellipse 5 → hyperbola.

② Circle i) $x^2+y^2=a^2 \rightarrow C=(0,0)$ $R=a$.

ii) $(x-a)^2+y^2=r^2$

iii) $x^2+(y-a)^2=r^2$

iv) $(x-a)^2+(y-b)^2=r^2$

v) $x^2+y^2+2gx+2fy+c=0$

③ Parabola $y^2=\pm 4ax$

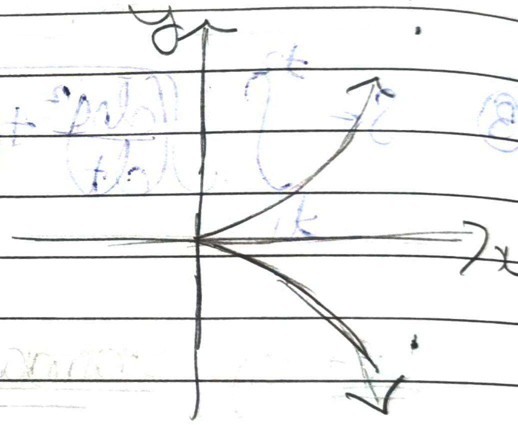
$x^2=\pm 4ay$

④ ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

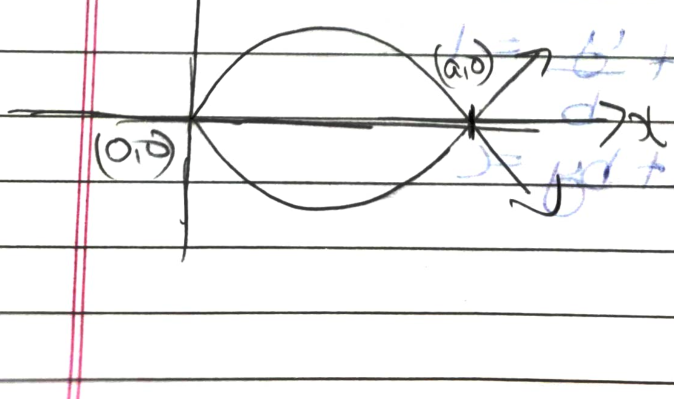
⑤ Hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

⑥ Semi-Cubical Parabola

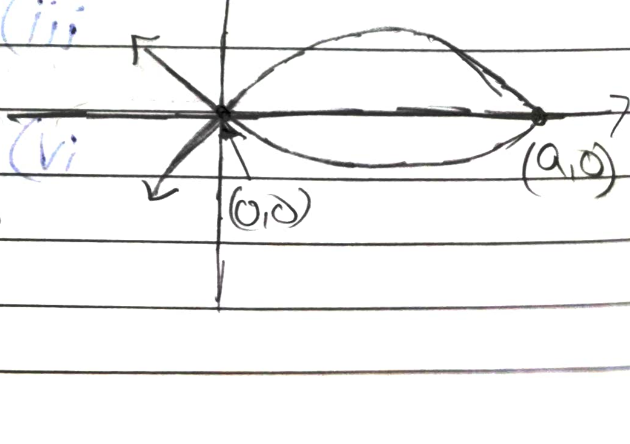
$ay^2=x^3$



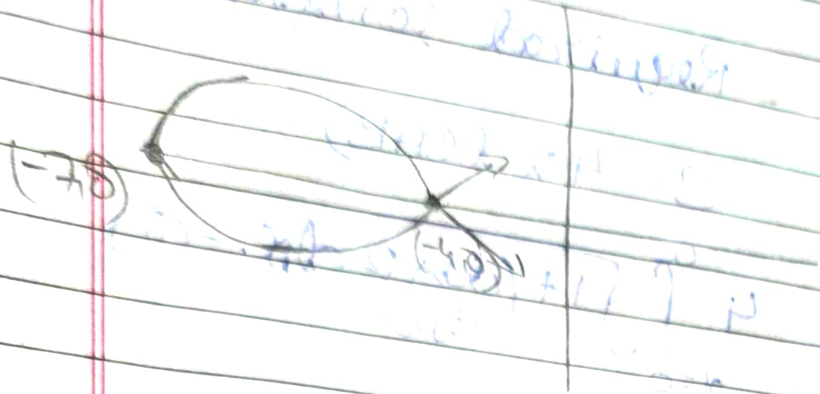
⑦ $3ay^2=x(a-x)^2$



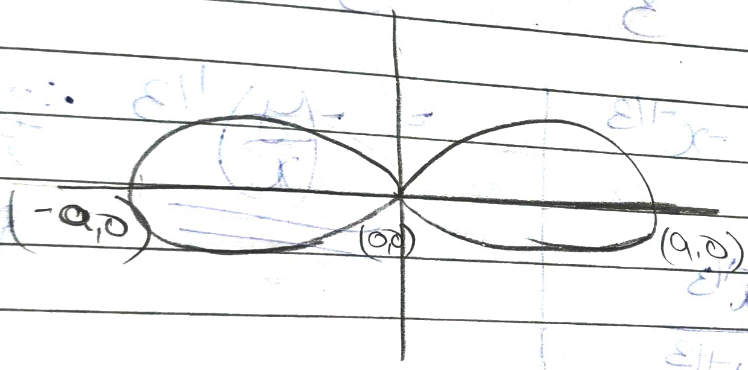
⑧ $3ay^2=x^2(a-x)$



(9). $9y^2 = (x+7)(x+4)^2$

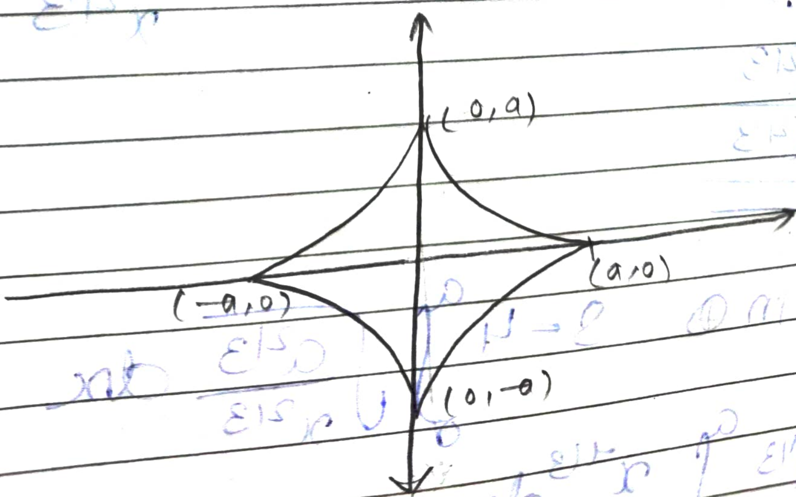


(10) $y^2 = x^2(a^2 - x^2)$ Bernoulli's of lemniscate



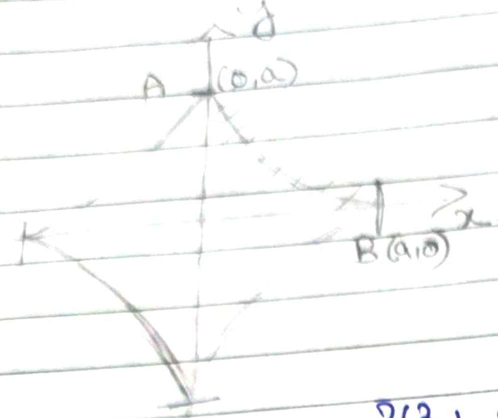
(11). Astroid Hypo cycloid

$x^{2/3} + y^{2/3} = a^{2/3}$



Q. Find the total length of $x^{2/3}+y^{2/3}=a^{2/3}$

Sol^n



Required length

$S = 4 \times l(AB)$

$$4\int_{x=0}^{a} \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx \quad \text{—(1)}$$

$$x^{2/3}+y^{2/3}=a^{2/3}$$

Diff wrt $x \rightarrow \frac{2}{3}x^{-1/3}+\frac{2}{3}y^{-1/3}\frac{dy}{dx}=0$

$$y^{-1/3}\frac{dy}{dx} = -x^{-1/3}$$

$$\frac{dy}{dx} = \frac{-x^{-1/3}}{y^{-1/3}}$$

$$= -\left(\frac{y}{x}\right)^{1/3}$$

$\because \frac{y}{x}$ to take the signature power.

$$1+\left(\frac{dy}{dx}\right)^2 = 1+\left(\frac{y}{x}\right)^{2/3} \rightarrow \frac{x^{2/3}+y^{2/3}}{x^{2/3}}$$

$$= \frac{a^{2/3}}{x^{2/3}}$$

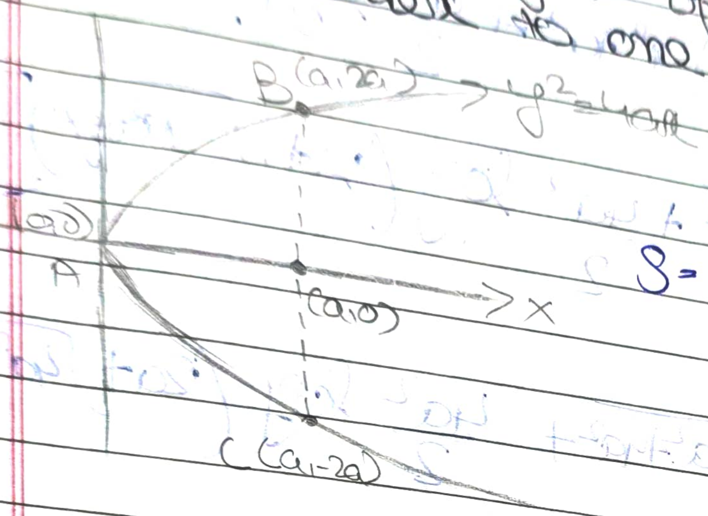
Subs in (1) $S = 4\int_0^a \sqrt{\frac{a^{2/3}}{x^{2/3}}}\, dx$

$$4a^{1/3}\int_0^a x^{-1/3}\, dx.$$

$$4a^{1/3}\left[\frac{x^{2/3}}{2/3}\right]_0^a \to 6a^{1/3}\times(a^{2/3}-0)$$

$$= \underline{\underline{6a}}$$

Q. Find the length of parabola $y^2=4ax$ from vertex to one end of latus rectum.



Req. length $S = l(AB)$

$$S=\int_0^a \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx \quad \text{①}$$

$$\int\sqrt{1+\frac{dx}{dy}}$$

Diff^n $y^2=4ax$ wrt $x \to 2y\frac{dy}{dx}=4a \to \frac{dy}{dx}=\frac{2a}{y}$

$$\left(\frac{dy}{dx}\right)^2=\frac{4a^2}{y^2} \to \frac{4a^2}{4ax}=\frac{a}{x}$$

Don't do with this formula, coz x is in denominator

Req. length $S = l(AB)$

$$S=\int_0^{2a}\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy \quad \text{①}$$

$$1+\left(\frac{dx}{dy}\right)^2 = 1+\frac{y^2}{4a^2}$$

Diff $y^2=4ax$ wrt y.

$$\to \frac{4a^2+y^2}{4a^2}$$

$$2y=4a\frac{dx}{dy}$$

$$\frac{dx}{dy}=\frac{y}{2a}$$

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Sub in ①.

$$S = \int_0^{2a} \frac{\sqrt{(2a)^2 + y^2}}{2a}\, dy$$

Formula $\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left(x+\sqrt{x^2+a^2}\right)$

$$S = \frac{1}{2a}\left[\frac{y}{2}\sqrt{4a^2+y^2} + \frac{4a^2}{2}\log\left(y+\sqrt{4a^2+y^2}\right)\right]_0^{2a}$$

$$S = \frac{1}{2a}\left\{\frac{2a}{2}\sqrt{4a^2+4a^2} + \frac{4a^2}{2}\log\left(2a+\sqrt{4a^2+4a^2}\right)\right.$$

$$\left. - \left[0 + \frac{4a^2}{2}\log\left(0+\sqrt{4a^2}\right]\right\}\right.$$

$$S = \frac{1}{2a}\left\{a(2\sqrt{2}a) + 2a^2\log(2a+2\sqrt{2}a) - 2a^2\log 2a\right\}$$

$$\frac{1}{2a}\left\{2\sqrt{2}a^2 + 2a^2\log\frac{2a(1+\sqrt{2})}{2a}\right\} \rightarrow \frac{2a^2}{2a}\left\{\sqrt{2}+\log(1+\sqrt{2})\right\}$$

$$\underline{a\left\{\sqrt{2}+\log(1+\sqrt{2})\right\}}$$

Find the length of Parabola $x^2=4y$ which lies inside the circle $x^2+y^2=6y$.

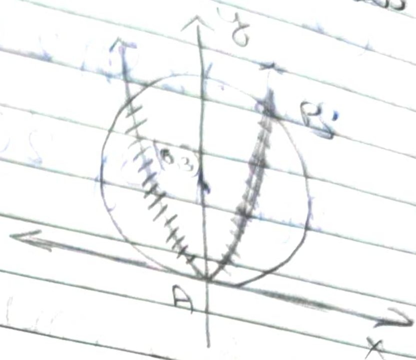
$x^2=4y$ — (1)

$x^2+y^2-6y=0$ — (2)

$x^2+y^2-6y+9-9$

$x^2+(y-3)^2=3^2$

$C=(0,3)$ $\underline{r=3}$



Reqd length $S=2\times l(AB)$

$C=(-g,-f)$
$r=\sqrt{g^2+f^2}$

$$S=2\int_{y_1}^{y_2}\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy \quad \text{— (3)}$$

Solving (1) & (2) $4y+y^2=6y \;\because (x^2=4y)$

$y^2-2y=0$

$y=0$, $y=2$ (1) → $y=0$ $x=0$

$y=2$ $\underline{x=2\sqrt{2}}$

$x^2=4y$ → diff wrt x. → $2x=4\frac{dy}{dx}$ → $\frac{dy}{dx}=\frac{x}{2}$

$1+\left(\frac{dy}{dx}\right)^2=1+\frac{x^2}{4} \to \frac{4+x^2}{4}$

Sub in (3) $S=2\int_0^{2\sqrt{2}}\sqrt{\frac{4+x^2}{4}}\,dx$

Formula: $\int\sqrt{x^2+a^2}\,dx=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\log\left(x+\sqrt{a^2+x^2}\right)$

$$S=\left[\frac{x}{2}\sqrt{x^2+4}+\frac{4}{2}\log\left(x+\sqrt{x^2+4}\right)\right]_0^{2\sqrt{2}}$$

∴ Don't expand

$$\left[\frac{2\sqrt{2}}{2}\sqrt{12} + 2\log(2\sqrt{2}+\sqrt{12})\right] - \left[0 + 2\log 2\right]$$

$$2\sqrt{6} + 2\log 2(\sqrt{2}+\sqrt{3}) - 2\log 2$$

$$2\left[\sqrt{6} + \log\frac{\cancel{2}(\sqrt{2}+\sqrt{3})}{\cancel{2}}\right] \to \underline{2\left[\sqrt{6}+\log(\sqrt{2}+\sqrt{3})\right]}$$

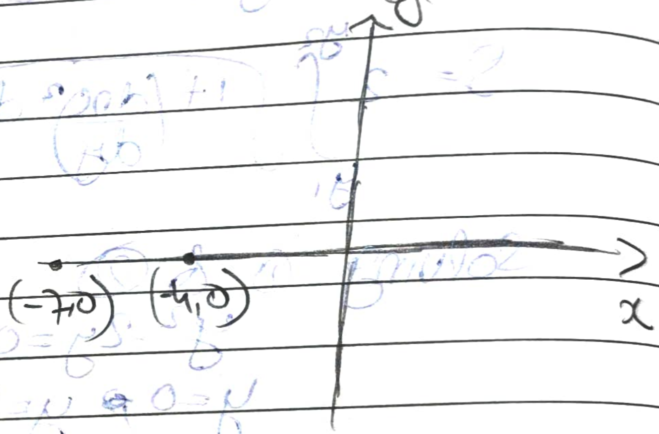
1. Find the perimeter of loop of the curve.

① $9y^2 = (x+7)(x+4)^2$ ② $3ay^2 = x(a-x)^2$
③ $3ay^2 = x^2(a-x)$

① $9y^2 = (x+7)(x+4)^2$

+ve

$y=0 \to x=-7,\ x=-4.$



$y^2 = + \to x+7>0.$
$\underline{x>-7}$

Reqd. length $S = 2 \times l(ABC)$

$$S = 2\int_{-7}^{-4}\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx \quad ①$$

$9y^2 = (x+7)(x+4)^2$
diff wrt x.
$18\frac{dy}{dx} = (x+7)2(x+4) + (x+4)^2$

$(x+4)[2x+14+x+4]$
$(x+4)(3x+18)$

$$\frac{dy}{dx} = \frac{\cancel{3}(x+4)(x+6)}{6\cancel{18}\,y} \to \frac{(x+4)(x+6)}{6y}$$

$$1+\left(\frac{dy}{dx}\right)^2 = \frac{1+(x+4)^2(x+6)^2}{36y^2}$$

$$1 + \frac{(x+4)^2(x+6)^2}{4\times(x+7)(x+4)^2}$$

$$\frac{4\cdot(x+7)+x^2+12x+36}{4(x+7)}$$

Sub in eqn ①.

$$S = 2\int_{-7}^{-4}\sqrt{\frac{(x+8)^2}{4(x+7)}}\,dx = \int_{-7}^{-4}\frac{x+8}{\sqrt{x+7}}\,dx.$$

$$= \int_{-7}^{-4}\frac{(x+7)+1}{\sqrt{x+7}}\cdot dx \rightarrow \int_{-7}^{-4}\frac{(x+7)+1}{\sqrt{x+7}}\,dx.$$

$$\rightarrow \int_{-7}^{-4}\left[\sqrt{x+7}+\frac{1}{\sqrt{x+7}}\right]\cdot dx$$

$$\int(ax+b)^n\cdot dx = \frac{(ax+b)^{n+1}}{a(n+1)}$$

$$S = \int_{-7}^{-4}(x+7)^{1/2}+(x+7)^{-1/2}\,]dx.$$

$$\left[\frac{(x+7)^{3/2}}{3/2}+\frac{(x+7)^{1/2}}{1/2}\right]_{-7}^{-4}$$

$$\frac{2}{3}(3)^{3/2}+2(3)^{1/2}-0 \rightarrow \underline{4\sqrt{3}}$$

no need to draw graph for starting & ending point it provided in ques.

- Prove that the length of the curve $y = \log\left(\frac{e^x - 1}{e^x + 1}\right)$ from $x=1$ to $x=2$ is $\log\left(e + \frac{1}{e}\right)$

Required length $S = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$

$$S = \int_1^2 \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \quad (1)$$

$$y = \log(e^x - 1) - \log(e^x + 1)$$

$$\frac{dy}{dx} = \frac{e^x}{e^x - 1} - \frac{e^x}{e^x + 1}$$

$$e^x \left[\frac{(\cancel{e^x}+1) - (\cancel{e^x}-1)}{e^{2x}-1}\right] \rightarrow \frac{dy}{dx} = \frac{2e^x}{e^{2x}-1}$$

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{4e^{2x}}{(e^{2x}-1)^2}$$

$(a-b)^2 + 4ab$
$\hookrightarrow (a+b)^2$

$$= \frac{(e^{2x}-1)^2 + 4e^{2x}}{(e^{2x}-1)^2} \rightarrow \frac{(e^{2x}+1)^2}{(e^{2x}-1)^2}$$

Subs in eq (1).

$$S = \int_1^2 \left(\frac{e^{2x}+1}{e^{2x}-1}\right)$$

$$S = \int_1^2 \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\int_1^2 \sqrt{\left(\frac{e^{2x}+1}{e^{2x}-1}\right)^2}$$

$$\frac{e^{2x}+1}{e^{2x}-1} = \frac{e^x(e^x+e^{-x})}{e^x(e^x-e^{-x})}$$

$$\int_1^2 \frac{e^{-x}(e^{2x}+1)}{e^{-x}(e^{2x}-1)}\,dx$$

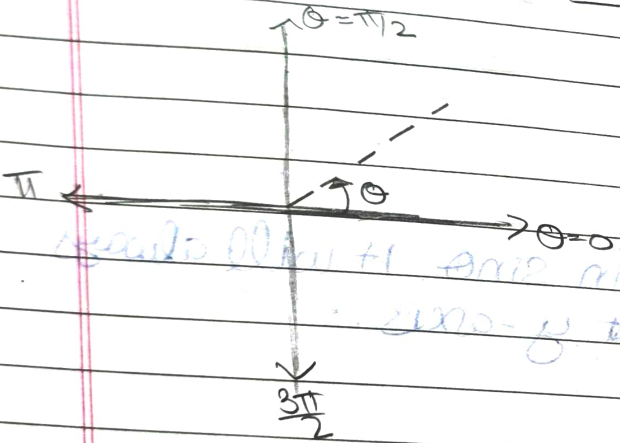
$$\rightarrow \int_1^2 \left(\frac{e^x+e^{-x}}{e^x-e^{-x}}\right)dx$$

$$\int \frac{f'(x)}{f(x)}\,dx = \log f(x)$$

$$\left[\log(e^x-e^{-x})\right]_1^2$$

$$-\log(e^2-e^{-2}) - \log(e-e^{-1})$$

Rectification in Polar Form



Standard Eqn's.

I) **Circle:**

1) $x^2+y^2=a^2$

put $x = r\cos\theta$

$y = r\sin\theta$.

$r^2 = a^2$

$\boxed{r=a}$ — (1).

$\because r=a \rightarrow$ radius so not negative

2) $(x-a)^2+y^2=a^2$

$x^2+y^2-2ax=0$.

$x^2+y^2=2ax$

$r^2=2ax$

$r^2=2a\,r\cos\theta$

$\boxed{r=2a\cos\theta}$ — (2).

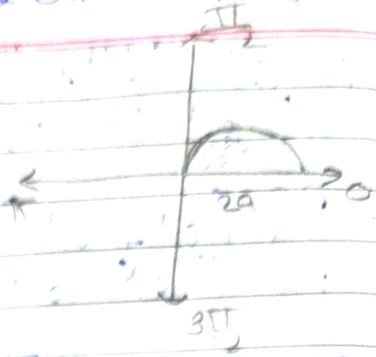
r ka value kbhi -ve nhi jana chahiye $\cos\theta$, $\sin\theta$ chalega

⭐ Jis jagah cosθ matlab circle's eqn.

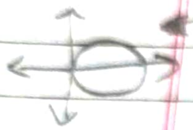
★ cosθ ka co-efficient will be diameter of circle [along x-axis]

$r = 2a\cos\theta$

θ	0	$\pi/2$
r	$2a$	0

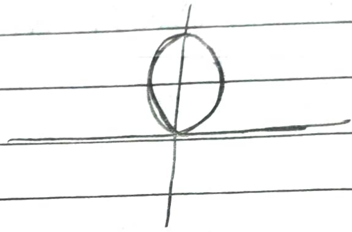


Rules: Whenever the curve is in cosθ, it will always be symmetrical about x-axis.



(3) $x^2 + (y-a)^2 = a^2$

$r = 2a\sin\theta$ —(3).



θ	0	$\pi/2$	π
r	0	$2a$	0

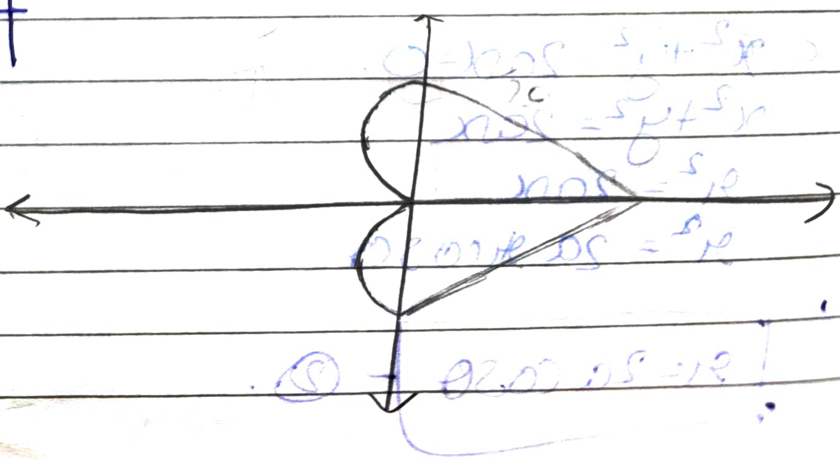
- Whenever the curve is in sinθ, it will always be symmetrical about y-axis.

(II) Cardioid's Eqn

1) $r = a(1+\cos\theta)$

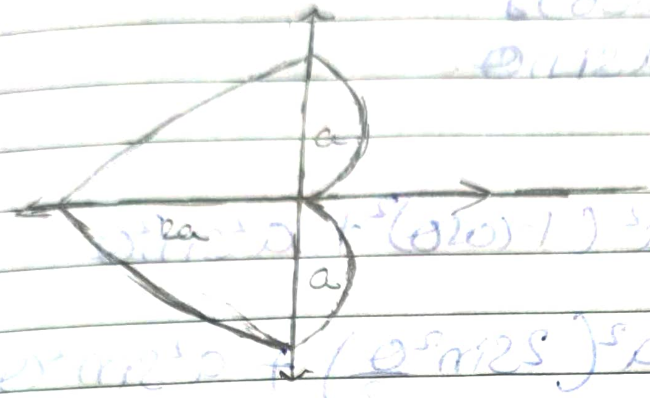
r=0 ~~tak~~ jaane ka

θ	0	$\pi/2$	π
r	$2a$	a	0



(2) $r = a(1-\cos\theta)$

θ	0	$\pi/2$	π	$3\pi/2$	2π
r	0	a	2a	a	2a



IMP

$\cos\theta \Rightarrow$ symmetrical about x-axis

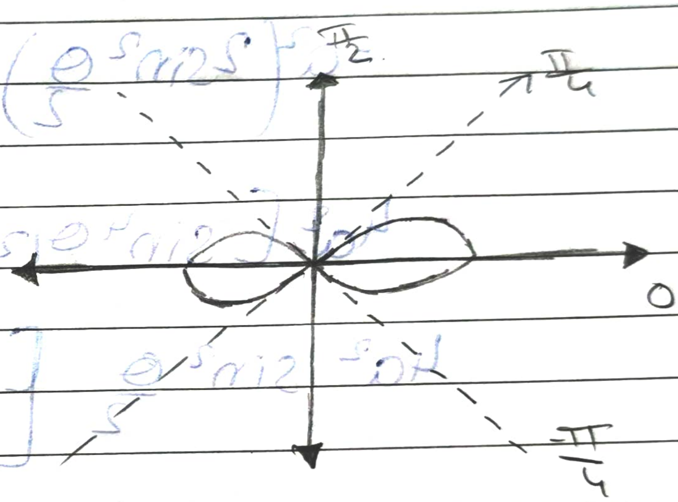
$\sin\theta \Rightarrow$ symmetrical about y-axis

$r^2 \Rightarrow$ symmetric about origin.

(3) III lemniscate.

$r^2 = a^2 \cos(2\theta)$

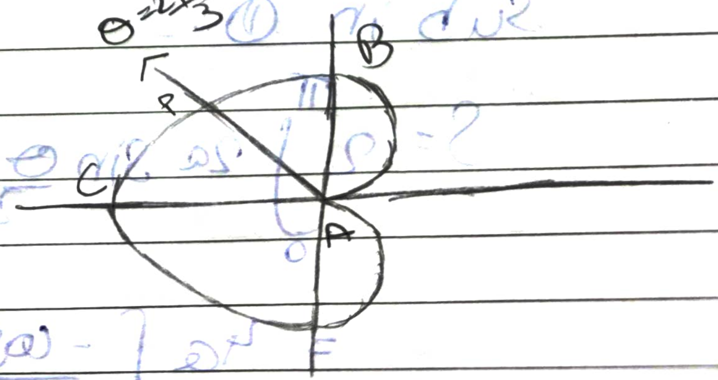
θ	0	$\pi/4$
r	a	0



Q. Find the perimeter of cardioid $\boxed{r = a(1-\cos\theta)}$ & ST the line $\theta = \frac{2\pi}{3}$ bisects the upper half of the cardioid.

$r = a(1-\cos\theta)$ —(1)

θ	0	$\pi/2$	π
r	0	a	2a



$1-\cos\theta = 2\sin^2\frac{\theta}{2}$

Reqd. length $S = 2\times l(ABC)$

$$S = 2\int_0^{\pi}\sqrt{r^2+\left(\frac{dr}{d\theta}\right)^2}\,d\theta \quad -(1)$$

$r = a(1-\cos\theta)$

$\frac{dr}{d\theta} = a\sin\theta$

$$r^2+\left(\frac{dr}{d\theta}\right)^2 = a^2(1-\cos\theta)^2 + a^2\sin^2\theta$$

$$= a^2\left(2\sin^2\frac{\theta}{2}\right) + a^2\sin^2\theta$$

$$= a^2\left(2\sin^2\frac{\theta}{2}\right) + a^2\left(2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)^2$$

$$4a^2\left[\sin^4\theta/2 + \sin^2\theta/2 \times \cos^2\theta/2\right]$$

$$4a^2\sin^2\frac{\theta}{2}\left[\underbrace{\sin^2\frac{\theta}{2} + 1\left(\cos^2\frac{\theta}{2}\right)}_{=1}\right]$$

$$4a^2\sin^2\frac{\theta}{2} \rightarrow \left(2a\sin\frac{\theta}{2}\right)^2$$

Sub in (1).

$$S = 2\int_0^{\pi} 2a\sin\frac{\theta}{2}\,d\theta$$

$$= 4a\left[\frac{-\cos\theta/2}{1/2}\right]_0^{\pi}$$

$$8a[0+1] = \underline{8a}$$

$$8a\left[-\cos\theta/2\right]_0^{\pi}$$

$$8a\left[(-0)+(1)\right] = \underline{8a}$$

We have to ST $l(AP) = 2a$.

$$l(AP) = \int_0^{2\pi/3} \sqrt{r^2+\left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

$$= \int_0^{2\pi/3} 2a\sin\frac{\theta}{2}\, d\theta$$

$$= 2a\left[\frac{-\cos\theta/2}{\theta/2}\right]_0^{2\pi/3} \rightarrow 4a\left[-\cos\theta/2\right]_0^{2\pi/3}$$

$$4a\left[-\cos\frac{\pi}{3}+\cos 0\right]$$

$$4a\left[-\frac{1}{2}+1\right] \rightarrow 4a\left(\frac{1}{2}\right) = \underline{2a}$$

The line $\theta = \frac{2\pi}{3}$ $\therefore$ bisects the upper half.

HW

1. ST the perimeter of $r^2 = a^2\cos 2\theta$ is $\frac{a}{\sqrt{2\pi}}\left(\Gamma\frac{1}{4}\right)^2$

2. To Find the length of cardioid $r = a(1-\cos\theta)$ which lies inside the circle $r = a\cos\theta$.

1. Find the length of cardioid $r = a(1-\cos\theta)$ which lies outside the circle $r = a\cos\theta$.

(heart) Cardioid $\rightarrow$ $r = a(1-\cos\theta)$ — (1)

(circle) $r = a\cos\theta$ — (2).

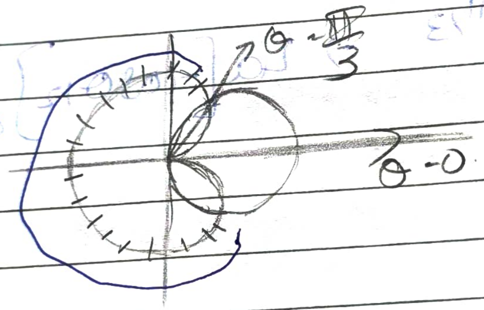
Solve (1) & (2).

$\cancel{a}(1-\cos\theta) = \cancel{a}\cos\theta$

$2\cos\theta = 1$

$\cos\theta = \frac{1}{2}$

$\theta = 60° \rightarrow \frac{\pi}{3}$



Reqd. length $S = 2 \times l(AB)$

$$S = 2\int_{\pi/3}^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta \quad (3)$$

$r = a(1-\cos\theta)$ consider this

$\cancel{r^2} + \left(\frac{dr}{d\theta}\right)^2$ $\quad \frac{dr}{d\theta} = a\sin\theta$

$$S = 2\int_{\pi/3}^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta \quad (3)$$

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$r^2 = (a^2 - a^2\cos^2\theta)$

$r^2 + \left(\frac{dr}{d\theta}\right)^2 = a^2[(1-\cos\theta)^2 + \sin^2\theta]$

$(1-\cos\theta)^2 = 1 - 2\cos\theta + \underbrace{\cos^2\theta + \sin^2\theta}_{\to 1}$

$2 - 2\cos\theta \to 2(1-\cos\theta)$

$\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} = a\sqrt{2(1-\cos\theta)} \to 1-\cos\theta = \left(2\sin^2\frac{\theta}{2}\right)$

$\sqrt{2(1-\cos\theta)} = \sqrt{4\sin^2\frac{\theta}{2}} = \underline{\underline{2\sin\frac{\theta}{2}}}$

$S = 2\int_{\pi/3}^{\pi} 2a\sin\frac{\theta}{2}\,d\theta. \to 4a\int_{\pi/3}^{\pi}\sin\frac{\theta}{2}\,d\theta.$

$u = \frac{\theta}{2} \to d\theta = 2du \quad \theta = \frac{\pi}{3} \to u = \frac{\pi}{6}$

$4a.\,2\int_{\pi/6}^{\pi/2}\sin u\,du \to 8a[-\cos u]_{\pi/6}^{\pi/2}.$

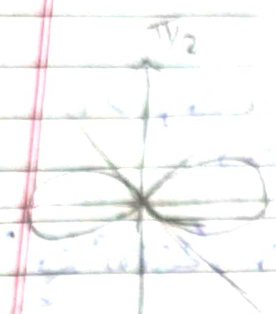
$\to 8a\left(-\cos\frac{\pi}{2} + \cos\frac{\pi}{6}\right) \to 8a\left(0 + \frac{\sqrt{3}}{2}\right)$

$\underline{\underline{4a\sqrt{3}}} \to \boxed{\underline{\underline{4\sqrt{3}a}}}$

length of cardioid ($r = a(1-\cos\theta)$) which is not a part of circle ($r = a\cos\theta$) is $\underline{\underline{4\sqrt{3}a}}$.

HW

Q. Show that the perimeter of lemniscate $r^2=a^2\cos(2\theta)$ is $\frac{a}{\sqrt{2\pi}}\left(\Gamma\frac{1}{4}\right)^2$



$$S = 4\int_0^{\pi/4} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

$$\cancel{2r}\quad r^2 = a^2\cos 2\theta$$

$$2r\,\frac{dr}{d\theta} = a^2(-2\sin 2\theta)$$

$$\cancel{\frac{d\theta}{dr} = \frac{a^2(-2\sin 2\theta)}{2r}} \qquad \frac{dr}{d\theta} = \frac{-a^2\sin 2\theta}{r}$$

$$\cancel{\frac{d\theta}{dr} = \frac{-a^2\sin 2\theta}{r}}$$

$$r^2 + \left(\frac{dr}{d\theta}\right)^2 = r^2 + \frac{a^4\sin^2 2\theta}{r^2} \rightarrow \frac{r^4 + a^4\sin^2 2\theta}{r^2}$$

Put $r^2 = a^2\cos 2\theta \rightarrow r^4 = a^4\cos^2 2\theta$

$$r^4 + a^4\sin^2 2\theta \rightarrow a^4\cos^2 2\theta + a^4\sin^2 2\theta$$

$$a^4(\underbrace{\cos^2 2\theta + \sin^2 2\theta}_{1}) \rightarrow a^4$$

$$r^2 + \left(\frac{dr}{d\theta}\right)^2 = \frac{a^4}{r^2}$$

$$\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} = \frac{a^2}{r}$$

Replace $r = a\sqrt{\cos 2\theta}$

$$\frac{a^2}{r} \rightarrow \frac{a^2}{a\sqrt{\cos 2\theta}} \rightarrow \frac{a}{\sqrt{\cos 2\theta}}$$

$$S = 4\int_0^{\pi/4} \frac{a}{\sqrt{\cos 2\theta}}\, d\theta$$

$$4a\int_0^{\pi/4} \frac{1}{\sqrt{\cos 2\theta}}$$

$2\theta = t \rightarrow d\theta = \frac{dt}{2}$ Limits $\theta \rightarrow 0$, $\pi/4 \rightarrow \pi/2$

$$S = 4a \cdot \frac{1}{2}\int_0^{\pi/2} \frac{dt}{\sqrt{\cos t}} \rightarrow 2a\int_0^{\pi/2} \cos^{-1/2} t \, dt.$$

$$\int_0^{\pi/2} \cos^m t \, dt = \frac{\Gamma\pi \, \Gamma\frac{m+1}{2}}{2\,\Gamma\left(\frac{m}{2}+1\right)}$$

$m = -1/2$ $\frac{m+1}{2} = \frac{1}{4}$

$\frac{m}{2}+1 = -\frac{1}{4}+1 \rightarrow \frac{3}{4}$

$$\rightarrow \frac{\sqrt{\pi}\,\Gamma\frac{1}{4}}{2\,\Gamma\frac{3}{4}} \times \frac{\Gamma\frac{1}{4}}{\Gamma\frac{1}{4}} \rightarrow \frac{\sqrt{\pi}}{2}\frac{\left(\Gamma\frac{1}{4}\right)^2}{\Gamma\frac{3}{4}\,\Gamma\frac{1}{4}}$$

$\Gamma(1-p)\,\Gamma p = \frac{\pi}{\sin p\pi}$.

$$\frac{\sqrt{\pi}}{2}\frac{\left(\Gamma\frac{1}{4}\right)^2}{\frac{\pi}{\sin\pi/4}} \rightarrow a\frac{\sqrt{\pi}}{2}\frac{\left(\Gamma\frac{1}{4}\right)^2}{\sqrt{2}\pi}.$$

$$\frac{a}{\sqrt{2\pi}}\left(\Gamma\frac{1}{4}\right)^2 \rightarrow \frac{a}{\sqrt{2\pi}}\left(\Gamma\frac{1}{4}\right)^2$$

Parametric Form

I) Cycloid

i) $x = a(\theta + \sin\theta)$
$y = a(1 + \cos\theta)$

ii) $x = a(\theta - \sin\theta)$
$y = a(1 - \cos\theta)$

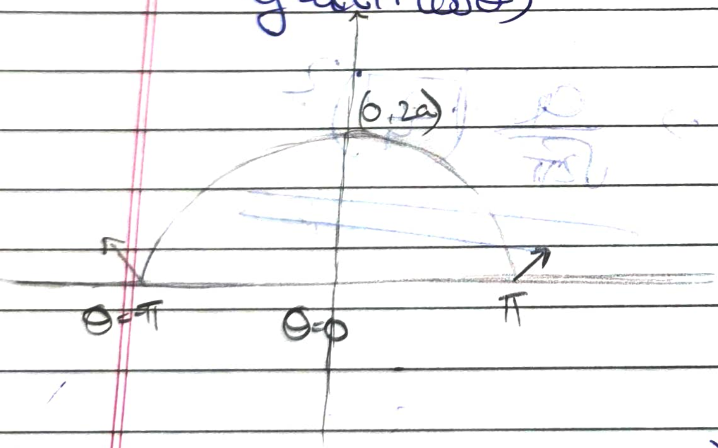
iii) $x = a(\theta + \sin\theta)$
$y = a(1 - \cos\theta)$

iv) $x = a(\theta - \sin\theta)$
$y = a(1 + \cos\theta)$

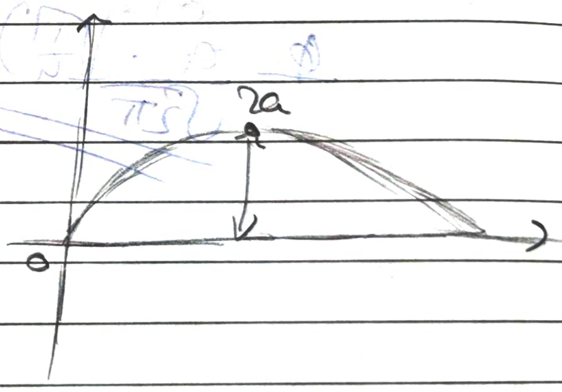
$2\pi \to$ Perimeter / circumference of circle.

cusp

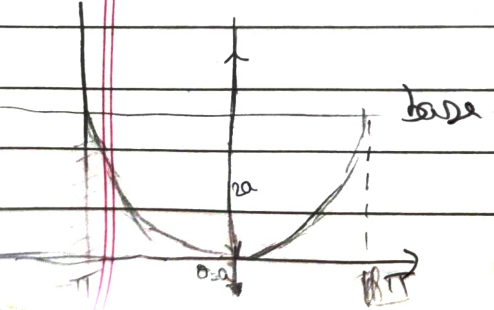
i) $x = a(\theta + \sin\theta)$
$y = a(1 + \cos\theta)$



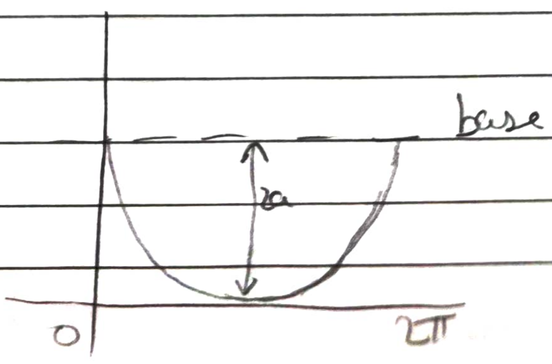
ii) $x = a(\theta - \sin\theta)$
$y = a(1 - \cos\theta)$



iii) $x = a(\theta + \sin\theta)$
$y = a(1 - \cos\theta)$



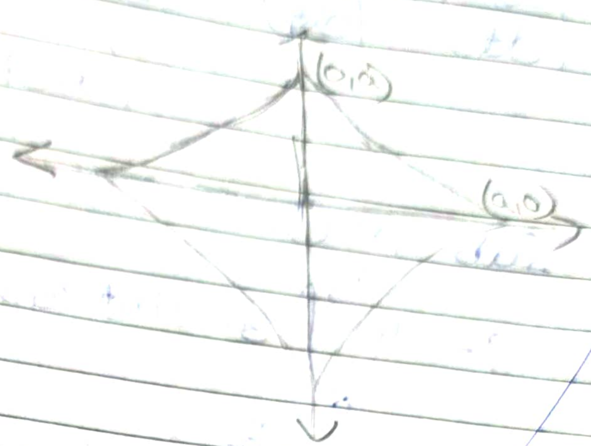
iv) $x = a(\theta - \sin\theta)$
$y = a(1 + \cos\theta)$



II) Astroid $(x^{2/3} + y^{2/3} = a)$

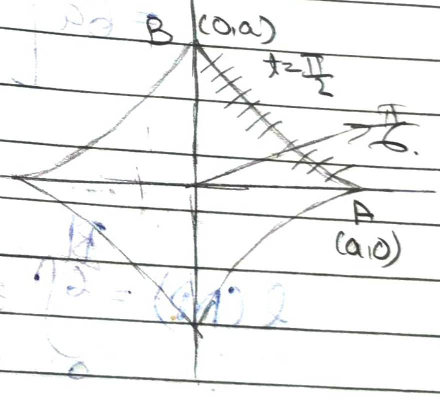
$x = a\cos^3 t$

$y = a\sin^3 t$



Q. Find the length of astroid $x = \cos^3 t,\ y = a\sin^3 t$. ST the line $\theta = \frac{\pi}{6}$ divides the 1^{st} quadrant in the ratio 1:3.

Reqd. length $= S = 4\, l(AB)$



$x = a \Rightarrow a = a\cos^3 t$

$\cos t = 1$

$\boxed{t = 0}$

$x = 0 \rightarrow a\cos^3 t = 0$

$\cos t = 0$

$\boxed{t = \frac{\pi}{2}}$

$x = a\cos^3 t$

$\frac{dx}{dt} = -3a\cos^2 t \sin t$

$y = a\sin^3 t$

$\frac{dy}{dt} = 3a\sin^2 t \cos t$

$x = a\cos^3 t$

$\frac{dx}{dt} = -3a\cos^2 t \sin t$

$y = a\sin^3 t$

$\frac{dy}{dt} = 3a\sin^2 t \cos t$

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9a^2\cos^4 t\,\sin^2 t + 9a^2\sin^4 t\cos^2 t$$

$$= 9a^2\cos^2 t\,\sin^2 t$$

$$= (3a\sin t\cos t)^2$$

Sub in ①.

$$S = 4\int_0^{\pi/2} 3a\sin t\cos t\,dt$$

$$= 6a\int_0^{\pi/2} (2\sin t\cos t)\,dt$$

$$= 6a\int_0^{\pi/2} \sin 2t\,dt$$

$$= 6a\left[-\frac{\cos 2t}{2}\right]_0^{\pi/2} \to 3a\left[-\cos\pi + \cos 0\right]$$

$$3a(1+1)$$

$$= \underline{6a}$$

$$l(AP) = \int_0^{\pi/6} 3a\sin t\cos t\,dt$$

$$\frac{3a}{2}\left[-\frac{\cos 2t}{2}\right]_0^{\pi/6} = \frac{3a}{4}\left[-\cos\frac{\pi}{3} + \cos 0\right]$$

$$\frac{3a}{4}\left[-\frac{1}{2}+1\right] = \underline{\frac{3a}{8}}$$

$$l(PB) = \int_{\pi/6}^{\pi/2} 3a \sin\theta \cos\theta \, d\theta.$$

$$= \frac{3a}{2}\left[\frac{-\cos 2\theta}{2}\right]_{\pi/6}^{\pi/2}.$$

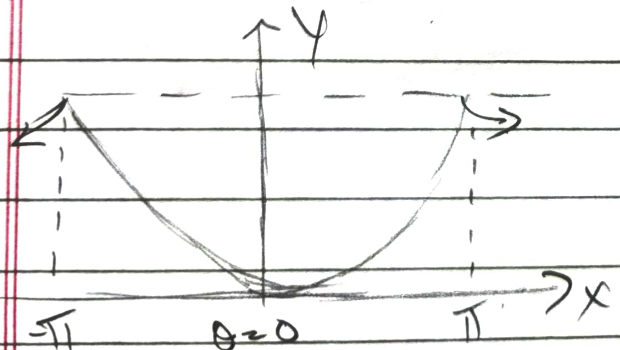
$$\frac{3a}{4}\left[1+\frac{1}{2}\right] \rightarrow \frac{9a}{8}.$$

$$\frac{l(AP)}{l(PB)} = \frac{1}{3}.$$

Q. Find the length of cycloid from one ~~centre~~ cusp to the next cusp.

$x = a(\theta + \sin\theta)$

$y = a(1-\cos\theta)$.



$S = 2 \times l(AB)$

$$= 2\int_0^{\pi} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\, d\theta. \quad (1)$$